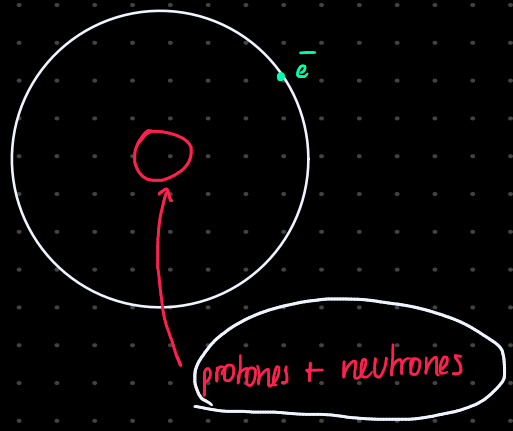
Astrofísica estelar II: procesos nucleares, evolución estelar y remanentes

1. Emisión nuclear

a. Emisión alfa (α)

$$ {}^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He $$

núcleo de helio-4



b. Emisión beta

$A = Z + n^o$

- beta menos β^-

$$ {}^{A}_{Z}X \longrightarrow {}^{A}_{Z+1}Y + e^- + \bar{\nu}_e $$

electrón — antineutrino electrón

- beta más β^+

$$ {}^{A}_{Z}X \longrightarrow {}^{A}_{Z-1}Y + e^+ + \nu_e $$

positrón — neutrino electrón

- captura de electrones

$$ {}^{A}_{Z}X + e^- \longrightarrow {}^{A}_{Z-1}Y + \nu_e $$

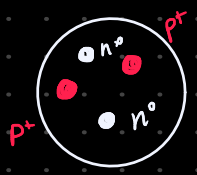
electrón — neutrino electrón

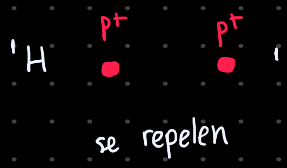
c. Emisión Gamma

$$ {}^{A}_{Z}X^{*} \longrightarrow {}^{A}_{Z}X + \gamma $$

fotón

2. Fusión nuclear





se repelen

$$ E_c = k\frac{ee}{r} = \frac{ke^2}{r} $$

→ necesitamos que se acerquen a 10^{-15} m ($1\ fm = 10^{-15}$ m)

efecto túnel

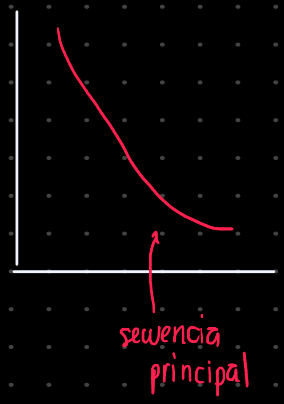
$$ K = \frac{3}{2} k_B T $$

movimiento

constante de Boltzmann $k_B = 1.38 \times 10^{-23}\ J K^{-1}$

a. Cadena protón - protón

$M \lesssim 1.3\, M_\odot$



PP-I

2 veces

① $^1H + {}^1H \longrightarrow {}^2H + \boxed{\nu_e} + 1.44\ MeV$

deuterio

② $^2H + {}^1H \longrightarrow {}^3He + \gamma + 5.49\ MeV$

③ $^3He + {}^3He \longrightarrow {}^4He + 2\,{}^1H + 12.86\ MeV$

→ PP-II (>14 millones K)

④ $^3He + {}^4He \longrightarrow {}^7Be + \gamma$

⑤ $^7Be + e^- \longrightarrow {}^7Li + \nu_e$

⑥ $^7Li + {}^1H \longrightarrow 2\,{}^4He$

→ PP-III (> 25 millones K) ~0.3 % en el Sol

④ $^7Be + {}^1H \longrightarrow {}^8B + \gamma$

⑤ $^8B \longrightarrow {}^8Be + e^+ + \nu_e$

⑥ $^8Be \longrightarrow 2\,{}^4He$

centrales en experimentos de neutrinos solares.

1 protón ≡ núcleo de hidrógeno

$p^+ \equiv {}^1H$

b. Ciclo CNO

$M > 1.3\, M_\odot$

① Carbono-12 captura un protón para formar nitrógeno-13

$${}^{12}_{6}C + {}^1H \longrightarrow {}^{13}_{7}N + \gamma$$

② Nitrógeno-13 es inestable y experimenta decaimiento beta +

$${}^{13}_{7}N \longrightarrow {}^{13}_{6}C + e^+ + \nu_e$$

③ Carbono-13 captura un protón y forma nitrógeno-14

$${}^{13}_{6}C + {}^1H \longrightarrow {}^{14}_{7}N + \gamma$$

④ Nitrógeno-14 captura un protón para formar oxígeno-15

$${}^{14}_{7}N + {}^1H \longrightarrow {}^{15}_{8}O + \gamma$$

(5) Oxígeno-15 es inestable y experimenta decaimiento beta positivo

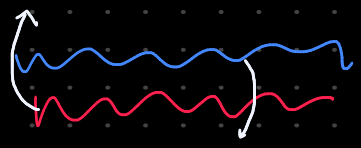
$$^{15}_{8}O \longrightarrow {}^{15}_{7}N + e^{+} + \nu_e$$

(6) Nitrógeno-15 captura un protón, emite una partícula alfa y regenera carbón-12

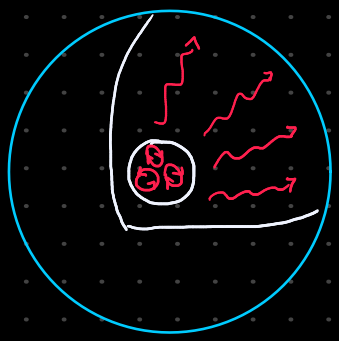
$$^{15}_{7}N + {}^{1}_{1}H \longrightarrow {}^{4}_{2}He + {}^{12}_{6}C$$

3. Transferencia de energía

opaco (convección) vs. transparente (radiación)



comp. convectivas



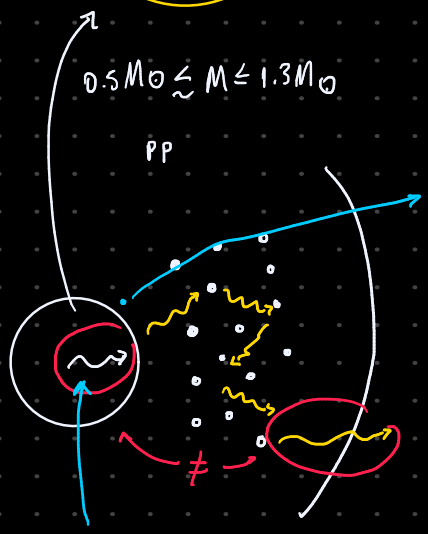
$M \gtrsim 1.3\,M_\odot$

CNO

$0.5\,M_\odot \lesssim M \lesssim 1.3\,M_\odot$

PP

$M \lesssim 0.5\,M_\odot$

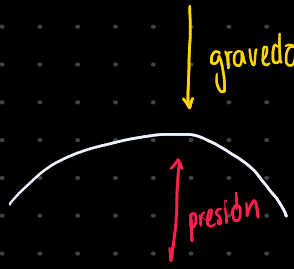


- ¿Cómo confirmamos que los procesos nucleares ocurren?

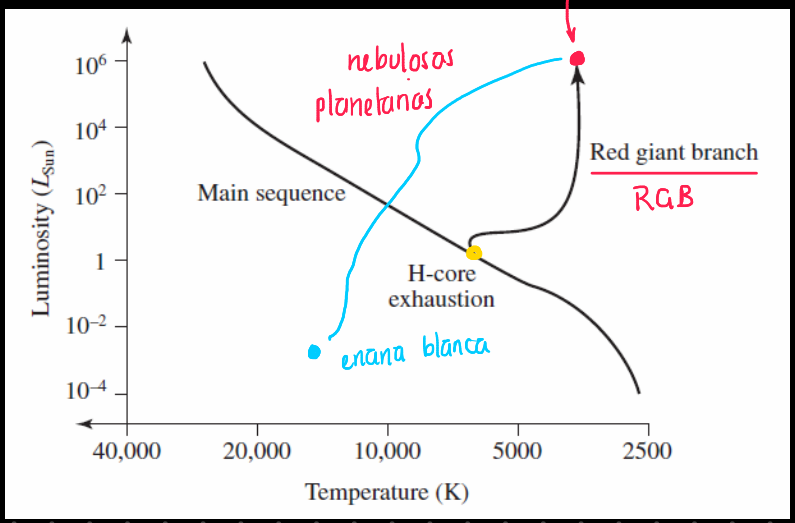
neutrinos

→ ¿Cómo se detectan los neutrinos?

4. Evolución de estrellas similares al Sol



equilibrio hidrostático.



$$\frac{L}{L_\odot} \approx 200\left(\frac{M_c}{0.3\,M_\odot}\right)^{7.6}$$

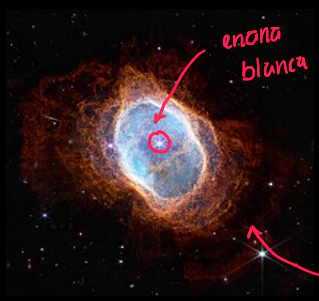
Helium flash.

→ Principio de exclusión de Pauli

s^1 ½ ↑ ↓ -½

↑↓

↑↑ ✗



5. Remanentes estelares

a. Enana blanca

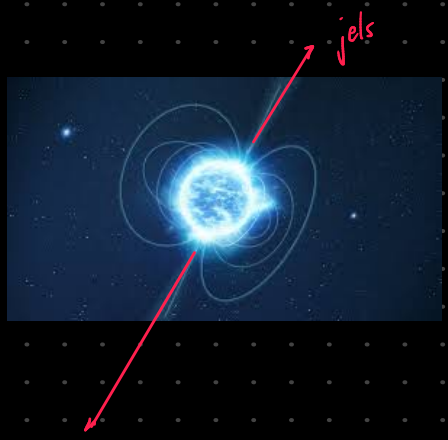
- $M < 8 M_\odot$

$$P_{grav} \sim \frac{G M^2}{R^4} \longrightarrow P_{deg} \sim \frac{\hbar^2 N^{5/3}}{m_e R^5}$$

$$n \sim \frac{N}{R^3} \qquad R \sim \frac{\hbar^2 N^{5/3}}{G m_e M^2} \qquad R \propto M^{-1/3}$$

Límite de Chandrasekhar

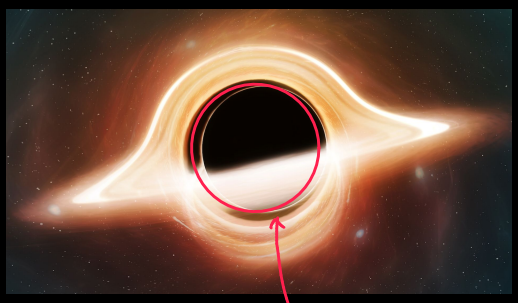
$$M_{ch} \sim \left(\frac{\hbar c}{G}\right)^{3/2} \frac{1}{m_p^2} \qquad \boxed{1.44 M_\odot}$$



b. Estrellas de neutrones

- captura de electrones $(p^+ + e^- \rightarrow n^0 + \nu_e)$

$\sim 10^{14}\ g\ cm^{-3}$



c. Agujero Negro

→ radio de Schwarzschild $\quad r_s = \frac{2GM}{c^2}$